Thermal Transport Through a Two-Layer Composite Rod

A synthetic scientific paper for the AgenQA public demo

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Abstract

We study steady one-dimensional heat conduction through a composite rod made of two homogeneous layers in perfect thermal contact. A closed-form resistance model gives the heat flux, interface temperature, and sensitivity to layer thickness. A small parameter study illustrates why the lower-conductivity layer dominates the total temperature drop. The model is intentionally compact but contains enough linked facts to support multi-step scientific reasoning tasks.

1. Physical model

The rod has cross-sectional area A and two layers arranged in series. Layer 1 has length $L1$ and thermal conductivity $k1$. Layer 2 has length $L2$ and thermal conductivity $k2$. The left boundary is held at temperature Th and the right boundary at Tc, with $Th > Tc$. Lateral heat loss and contact resistance are neglected, and all material properties are constant.

For steady one-dimensional conduction, each layer has thermal resistance

$$R1 = L1 / (k1\ A),\ R2 = L2 / (k2\ A).$$

The heat rate is

$$Q = (Th - Tc) / (R1 + R2),$$

and the heat flux is $q = Q / A$. Because the same heat rate crosses both layers, the interface temperature Ti can be written in either equivalent form:

$$Ti = Th - Q\ R1 = Tc + Q\ R2.$$

These equations also imply that the fraction of the total temperature drop occurring in layer 2 is $R2 / (R1 + R2)$.

2. Reference configuration

The reference rod uses A = 1.00e-2 m^2, $L1$ = 2.00e-2 m, $k1$ = 20 W m^-1 K^-1, $L2$ = 3.00e-2 m, and $k2$ = 5 W m^-1 K^-1. Boundary temperatures are Th = 400 K and Tc = 300 K.

For this configuration, $R1$ = 0.10 K W^-1 and $R2$ = 0.60 K W^-1. Therefore the total resistance is 0.70 K W^-1, the heat rate is 142.857 W, and the heat flux is 1.42857e4 W m^-2. The interface temperature is 385.714 K. Layer 2 accounts for 6/7 of the total temperature drop, even though it contains only 3/5 of the total rod length.

3. Thickness perturbation

We next double the thickness of layer 2 while holding A, $k1$, $k2$, Th, and Tc fixed. The new layer-2 resistance is $R2'$ = 1.20 K W^-1 and the total resistance is 1.30 K W^-1. The new heat rate is Q' = 76.923 W. The ratio Q'/Q is 7/13, showing that doubling one layer's thickness does not simply halve the heat rate when another series resistance remains present.

The perturbed interface temperature is $Ti' = Th - Q' R1$ = 392.308 K. The interface becomes hotter because the enlarged downstream resistance causes a larger fraction of the total temperature drop to occur in layer 2.

4. Discussion

The series-resistance formulation exposes a short dependency chain. Geometry and conductivity determine $R1$ and $R2$; the resistances determine Q; and Q with either layer resistance determines Ti. A solver that is shown only the original parameters and asked for the perturbed interface temperature must reconstruct all intermediate quantities. Locally, however, every transition can be checked using a single equation. This separation between local verification and global reconstruction makes the example suitable for demonstrating Edge and Path views.

5. Limitations

The model assumes constant conductivity, perfect thermal contact, and no lateral losses. It should not be used to predict a real composite without checking those assumptions. All values in this paper are synthetic and included only for software demonstration.